VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Experiment No. 04

Aim: - Write a Java Program on:

a) use of single Dimensional array.
b) use of multidimensional array

Software Requirement: - Java Development Kit (JDK), Integrated Development Environment (IDE) (NetBeans), Java Runtime Environment (JRE).

Theory: -

Java Arrays

An array is typically a grouping of elements of the same kind that are stored in a single, contiguous block of memory.

Java array is an object which contains elements of a similar data type. Additionally, the elements of an array are stored in a contiguous memory location. It is a data structure where we store similar elements. We can store only a fixed set of elements in a Java array.

Array in Java is index-based, the first element of the array is stored at the 0th index, 2nd element is stored on 1st index and so on.

Types of Array in Java

There are two types of arrays in Java:

1. Single-dimensional array in Java
2. Multi-dimensional array in Java

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1. Single-Dimensional Array in Java

- A 1D Array in Java is a linear array that allows the storage of multiple values of the same data type.
- It's a collection of data that stores elements of the same type in a sequentially allocated space in memory.
- Single-dimensional arrays can be utilized to store both simple and complex data types, anything from strings, integers, and booleans to custom-made classes depending on the user's requirements.
- In memory, a one or single-dimensional array holds the data in a linear list.
- The index of the memory runs from the array size of 0 to -1.
- Single-Dimensional Arrays are initialized just like regular variables, but instead of assigning a single value the programmers can pass several values separated by commas or define an array with static size.
- 1D Arrays are one of the handy tools contained within Java.
- They are useful for organizing data quickly and efficiently as well as giving us access to stored data using looping structures.

Syntax:

arrayName = new DataTye[length 1][length 2]....[length N];

OR

DataType[1st Dimension][2nd Dimension]....[Nth Dimension] arrayName = new DataTye[length 1][length 2]....[length N];

Example:

```
//declaration
int iArray1[] = new int[4];

//initialization
iArray1[0] = 32;
iArray1[1] = 13;
iArray1[2] = 56;
iArray1[3] = 78;
```

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2. Multi-Dimensional Array in Java

- Multi-Dimensional Array in Java is an array of arrays, i.e. it is an array object that has multiple dimensions.
- Multi-dimensional arrays are useful when dealing with a large amount of data since they give the ability to store and access data from a single variable but with multiple levels of hierarchy.
- This multi-dimensional array can be expanded to a certain number of dimensions such as two dimensions, three dimensions, etc.
- Multi-dimensional arrays in Java can hold any type of item, i.e. two-dimensional arrays can hold int values or objects such as strings and objects.
- Multi-dimensional arrays also have several methods available to help search and arrange the data within the array, making them very flexible and efficient when dealing with complex tasks.

Syntax:

Datatype [][]Array_Name = new Datatype[row_size][col_size(optional)];

OR

Datatype Array_Name[] = [{Value separated by comma},{Value separated by comma},....}];

Example:

```
//declaration
int matrix[] = new int[2][2];

//initialization
matrix[0][0] = 1;
matrix[0][1] = 2;
matrix[1][0] = 3;
matrix[1][1] = 4;
```

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Programs and Output:

a) Single Dimensional Array

```
public class SingleDimensionalArray {
    public static void main(String[] args) {
        // Declare and initialize a single-dimensional array
        int[] numbers = {10, 20, 30, 40, 50};

        // Print elements of the array
        System.out.println("Elements in the single-dimensional array:");
        for (int i = 0; i < numbers.length; i++) {
            System.out.println("Element at index " + i + ": " + numbers[i]);
        }
    }
}
```

Output:

```
Elements in the single-dimensional array:
Element at index 0: 10
Element at index 1: 20
Element at index 2: 30
Element at index 3: 40
Element at index 4: 50
```

b) Multi-Dimensional Array

```
public class MultiDimensionalArray {
    public static void main(String[] args) {
        // Declare and initialize a 2x2 two-dimensional array
        int[][] matrix = {
            {1, 2},
            {3, 4}
        };

        // Print elements of the two-dimensional array
        System.out.println("Elements in the two-dimensional array:");
        for (int i = 0; i < matrix.length; i++) {
            for (int j = 0; j < matrix[i].length; j++) {
```

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```
            System.out.println("Element at [" + i + "][" + j + "]: " + matrix[i][j]);
        }
    }
  }
}
```

Output:

```
Elements in the two-dimensional array:
Element at [0][0]: 1
Element at [0][1]: 2
Element at [1][0]: 3
Element at [1][1]: 4
```

Conclusion: Hence, in this practical, we learned and executed the Single Dimensional Array and Multi-Dimensional Array in Java Successfully.